



Features of special number sequences

1 Which of these are square numbers? Tick **3** correct answers

☒ 1

☐ 2

☐ 10

☒ 49

☒ 100

2 Which of these arrays represent a square number? Tick **1** correct answer


☐
☐
☐
☒

3 Which of these are cube numbers? Tick **2** correct answers

☒ 1

☐ 3

☐ 9

☐ 16

☒ 27



4 Match each sequence with its term-to-term rule. Write the correct letter in each box

a	4, 12, 36, 108, ...
b	4, 12, 20, 28,
c	4, 12, 22, 34, ...
d	4, 12, 21, 31, ...
e	4, 12, 28, 60, ...

a	multiply previous term by 3 to get the next term
c	add 8, then add 10, increasing the difference by 2 each time
d	add 8, then add 9, increasing the difference by 1 each time
b	add 8 to the previous term to get the next term
e	multiply the previous term by 2 and add 4 to get the next term

5 Which of these could be a position-to-term rule for the sequence which starts 5, 8, 11, 14, 17, ... ? Tick 1 correct answer

- ☐ add 3 to the term number
- ☐ multiply the term number by 2 and add 3
- ☐ multiply the term number by 3
- ☒ multiply the term number by 3 and add 2
- ☐ multiply the term number by 4

6 Which of these could be the first 4 terms of a geometric sequence? Tick 2 correct answers

- ☐ 1, 4, 8, 16
- ☒ 8, 12, 18, 27
- ☐ 5, 15, 25, 35
- ☒ 125, 25, 5, 1
- ☐ 0.5, 1, 2, 3.5

